

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

May – June 2015

PH121 Pharmaceutical Biology-II

Date: 13/05/2015, Wednesday Time: 10.00 a.m. to 1.00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q.1 Attempt all questions. Each question is of one mark.

20

1. One of the following is not a part of lymphatic organ or tissue
 - A. Liver
 - B. Spleen
 - C. Thymus
 - D. Lymph nodule
2. Anatomical fetures of Lymph nodes consists of
 - A. Cells of medulla & cortex
 - B. Cells around germinal center
 - C. Cells of hypothalamus
 - D. Both A & B
3. One of the Following is the correct sequence of flow of Lymph in the body
 - A. Blood → Interstitial fluid → Lymph vessel, duct → Jugular & subclavian vein
 - B. Blood → Lymph vessel, duct → Interstitial fluid → Jugular & subclavian vein
 - C. Lymph vessel, duct → Blood → Interstitial fluid → Jugular & subclavian vein
 - D. Interstitial fluid → Lymph vessel, duct → Blood → Jugular & subclavian vein
4. Lysozyme is present in which tissue / fluid?
 - A. Urine
 - B. Blood
 - C. Breast Milk
 - D. Lacrimal tissue

P.T.O

5. Which of the following is a Pentamer antibody?
 - A. Ig A
 - B. Ig M
 - C. Ig G
 - D. Ig D
6. Following is not an antimicrobial substance present in the body
 - A. Dermidine
 - B. Defensin
 - C. Cytokine
 - D. Thrombocidin
7. What is the purpose of RNA polymerase in Protein synthesis
 - A. For Translation
 - B. Catalysis of Transcription of DNA
 - C. Releasing m-RNA molecules
 - D. Detachment of t-RNA
8. What is a communicable disease?
 - A. Disease that has a long duration and is not easily cured.
 - B. Disease that is publically and scientifically understood
 - C. Disease that is related to eating and drinking
 - D. Disease that can be spread from one person to another.
9. Which of the following is not a communicable disease?
 - A. Autoimmune disease
 - B. Diphtheria
 - C. Typhoid
 - D. Malaria
10. Which of the following pair of disease is caused by Virus?
 - A. Rabies, Mumps
 - B. AIDS, Syphilis
 - C. Typhoid, Tetanus
 - E. Cholera, Tuberculosis
11. Following one of the technique is not for transferring Embryonic stem cells is
 - A. Invitro fertilization
 - B. Nuclear transfer
 - C. Sterilization
 - D. DNA finger printing
12. Transgenic mice model for Alzheimers disease expresses genes as
 - A. Protein kinase gene
 - B. ~~Rennin & angiotensin gene~~
 - C. APOE4 & Preselin 1 Gene
 - D. None of above
13. Embryonic stem cells are derived from which cell type?
 - A. From any host cell
 - B. From inner cellmass of adult cell
 - C. From adult cells
 - D. Inner cell mass of blastocyte

P.T.O

14. Normal blood plasma Ca^{++} concentration is
- A. 50-60 mEq/lit
 - B. 10-11 mEq/lit
 - C. 4.5-5.5 mEq/lit
 - D. 1.7-2.1 mEq/lit
15. Following are the sign & symptoms of Hyperkalemia
- A. Delirium
 - B. Muscle fatigue, mental confusion, Rise in urine output, Change in ECG
 - C. Numbness, tingling, hyperactive reflexes, muscle cramp
 - D. Muscular weakness, diarrhoea, cause of death by Ventricular fibrillation
16. Respiratory acidosis defines as
- A. Abnormally high Pco_2 in systemic arterial blood above 45 mmHg
 - B. Abnormally low Pco_2 in systemic arterial blood above 30 mmHg
 - C. Abnormally high Po_2 in systemic arterial blood above 100 mmHg
 - D. Abnormally high Po_2 in systemic arterial blood below 45 mmHg
17. Blood Calcium level is controlled by which hormones?
- A. ADH
 - B. Parathyroid hormone
 - C. Thyroid Hormone
 - D. Calcitriol
18. Followings vitamins are considered as Antioxidant Vitamins
- A. Vit. A,D,E
 - B. Biotin, Thiamin
 - C. Vit. C,E
 - D. Vit. B12
19. Which one of these vitamins may mask the anaemia of vitamin B12 deficiency?
- A. Biotin
 - B. Folic acid
 - C. Riboflavin
 - D. Thiamin
20. Which of the following type of RNA directs protein synthesis?
- A. t-RNA
 - B. r-RNA
 - C. m-RNA
 - D. s-RNA

P.T.O

SECTION – II

Q.2 Attempt any **FOUR** of the following

- | | | |
|---|---|----|
| A | Provide detail composition of Lymph with their Physiological role in Human body | 05 |
| B | Explain in short about Lymphatic organs & tissue | 05 |
| C | Explain in brief about formation & flow of Lymph in the body (Explain by chart) | 05 |
| D | Explain concept of regulations of gene Expression | 05 |
| E | Write a short note on DNA & Its replication Process | 05 |
| F | Write a short note on RNA & Translation process | 05 |

SECTION – III

Q.3 Attempt any **FOUR** of the following

- | | | |
|---|--|----|
| A | What is immunity? Describe the innate immunity in details giving examples. | 05 |
| B | Explain Immune interactions in detail with examples | 05 |
| C | Answer the following: | |
| | a) Differentiate between active and passive immunity | 03 |
| | b) Define Epidemic, Pandemic condition | 02 |
| D | Define recombinant DNA technology. Describe its applications in detail. | 05 |
| E | Explain different types of viral vectors in detail. | 05 |
| F | Describe the types of Stem cells in brief and provide its applications | 05 |

Q.4 Attempt any **FOUR** of the following

- | | | |
|---|---|----|
| A | Explain the following terms: | 05 |
| | 1) Tuberculosis 2) AIDS 3) Typhoid 4) Antibodies 5) Malaria | |
| B | Explain followings in Brief: | |
| | a) Application of transgenic animals. | 03 |
| | b) Applications of biotechnology in molecular diagnostics in brief. | 02 |
| C | Explain the physiological role and diseases related to Calcium or Sodium. | 05 |
| D | Enlist Immune system Disorders. Explain Antimicrobial substance in short | 05 |
| E | Classify Vitamins. Write short note on vitamin C or Vitamin B complex. | 05 |
| F | Explain functions and deficiency conditions of iron and iodine. | 05 |